

Karim Kabbara

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EXPERIENCE

Software Engineer

September 2023 – Present

Geotech Environmental Equipment, Inc.

Denver, CO

- Lead the full-stack development of Geotech's customer portal using React and Losant, enabling customers to remotely monitor and control 40+ field devices in real time.
- Manage ingestion and storage of 75,000+ data points per day using a tiered storage strategy with a time-series database (hot), BigQuery (warm), and Google Cloud Storage (cold), automating daily migration via Cloud Functions to optimize query performance and reduce long-term storage costs.
- Implement a highly customizable alert system enabling customers to respond to device failures and proactively schedule preventative maintenance, with a 90%+ adoption rate.
- Design a role-based access control system allowing for different user classes with scoped permissions across devices and admin-level control, with Vitest unit and integration tests covering critical paths.
- Program fault-tolerant, autonomous embedded systems with Node.js (Node-RED) and C and assist in the design of UL 508A certified control panels for environmental remediation applications, in collaboration with mechanical and electrical engineers.

Lead Engineer

August 2024 – Present

RealeContracts LLC

Remote

- Collaborate with a non-technical founder to build a real estate transaction platform on Google Cloud Platform targeting Colorado's 20,000+ brokers, leading all technical architecture and development.
- Architect a scalable containerized full-stack platform on Cloud Run, with Next.js and Express.js services, Prisma ORM, and Cloud SQL for PostgreSQL, with Firebase Auth securing broker and client access via JWT and role-based access control.
- Implement a least-privilege IAM policy by provisioning service accounts per service, granting only the necessary permissions to reduce attack vectors and blast radius.
- Build a server-side PDF generation service using Playwright and Cloud Functions that renders 20-page contracts in less than 3 seconds.
- Engineer a secure digital signature system with AES-256-GCM encryption and RSA key wrapping, supporting up to 16 multi-party signers per contract, backed by 20+ Vitest unit and integration tests covering encryption, signing, and data transport integrity.
- Transform 70+ state-provided PDF contracts into dynamic React forms with built-in signature capture, deadline tracking, error detection, and contract-aware AI assistance powered by Gemini, reducing human error and accelerating transaction closings.

EDUCATION & CERTIFICATIONS

Bachelor of Science in Computer Science

University of Colorado, Denver | GPA: 3.56

Cisco Certified Network Associate

CompTIA Security+ (SY0-601)

SKILLS

Frontend: TypeScript, React, Next.js, Tailwind, Tanstack Query, Zustand

Backend: Node.js, Express.js, REST APIs, PostgreSQL, Prisma ORM, Vitest, Playwright

Cloud & Infrastructure: Google Cloud Platform (Cloud Run, Cloud Functions, Cloud SQL, Cloud Storage, BigQuery, IAM, Secret Manager), Docker, Firebase, Vercel AI SDK, Gemini, Git